Description

Image





Caption

1. Gas turbine. © Kawasaki Turbines 2. Single superalloy blade. © Kawasaki Turbines

The material

With a name like "superalloy" there has to be something special here. There is. Superalloy is a name applied to nickel-based, iron-based and cobalt-based alloys that combine exceptional high-temperature strength with excellent corrosion and oxidation resistance. Without them, jet engines would not be practical: they can carry load continuously at temperatures up to 1200 C. The nickel-based superalloys are the ultimate metallic cocktail: nickel with a good slug of chromium and lesser shots of cobalt, aluminum, titanium, molybdenum, zirconium and iron. The data in this record span the range of high-performance nickel-based superalloys.

Compositional summary

Ni + 10 to 25% Cr + Ti, Al, Co, Mo, Zr, B and Fe in varying proportions.

General properties

Density		7.88e3	-	8.68e3	kg/m^3
Price	*	18.3	-	21.4	GBP/kg
Date first used		1944			

Mechanical properties

Young's modulus		200	-	220	GPa
Shear modulus	*	73	-	86	GPa
Bulk modulus	*	150	-	200	GPa
Poisson's ratio	*	0.28	-	0.33	
Yield strength (elastic limit)		273	-	900	MPa
Tensile strength		630	-	1.18e3	MPa
Compressive strength	*	280	-	910	MPa

Elongation		8.6	-	60	% strain
Hardness - Vickers		160	-	400	HV
Fatigue strength at 10^7 cycles	*	263	-	625	MPa
Fracture toughness	*	127	-	251	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	2e-4	-	7e-4	

Thermal properties

Melting point		1.26e3	-	1.4e3	°C
Maximum service temperature	*	777	-	1.04e3	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Poor conductor			
Thermal conductivity		10	-	14.6	W/m.°C
Specific heat capacity		400	-	480	J/kg.°C
Thermal expansion coefficient		10.5	-	14.3	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Poor conductor			
Electrical resistivity		98	-	140	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		3			
Formability		3	-	4	
Machinability		3			
Weldability		4	-	5	
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Acceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Acceptable
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent

Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Acceptable
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Excellent
Tolerance above 850 C (1562 F)	Excellent

Geo-economic data for principal component

Annual world production, principal component	2.37e6	tonne/yr
Reserves, principal component	1.05e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 17.1	-	18.9	kg/kg
Embodied energy (virgin grade production)	* 325	-	358	MJ/kg
Water usage	* 328	-	362	l/kg

Material processing: energy

Metal casting, embodied energy	* 46	-	52.8	MJ/kg
Metal extrusion, hot, embodied energy	* 22.3	-	29.9	MJ/kg
Metal rolling and forging, embodied energy	* 11.7	-	15.5	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg

Metal powder forming, embodied energy	* 58.4	-	69.5	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1.02e4	-	1.23e4	MJ/kg
Machining, coarse, embodied energy	* 17.8	-	20.5	MJ/kg
Machining, fine, embodied energy	* 24.3	-	29.2	MJ/kg
Machining, grinding, embodied energy	* 30.7	-	37.9	MJ/kg
Machining, laser, CO2, embodied energy	* 1.21	-	1.46	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.29	-	3.8	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.46	-	1.96	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.763	-	1.02	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 4.76	-	5.67	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 839	-	1.01e3	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.17	-	1.34	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.59	-	1.91	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.01	-	2.48	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0998	-	0.12	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 2.22	-	2.46	kg/kg
Embodied energy, recycling	* 22.8	-	25.1	MJ/kg
Recycle fraction in current supply	38			%
Combust for energy recovery	✗			
Landfill	✗			
Biodegrade	✗			
Toxicity rating	Slightly toxic			
A renewable resource?	✗			

Environmental notes

About 10% of the population is sensitive to nickel, causing them to react even to the nickel in stainless steel watch straps. Compounds of nickel can be more toxic; nickel carbonyl, used in the extraction of nickel, is deadly.

Supporting information

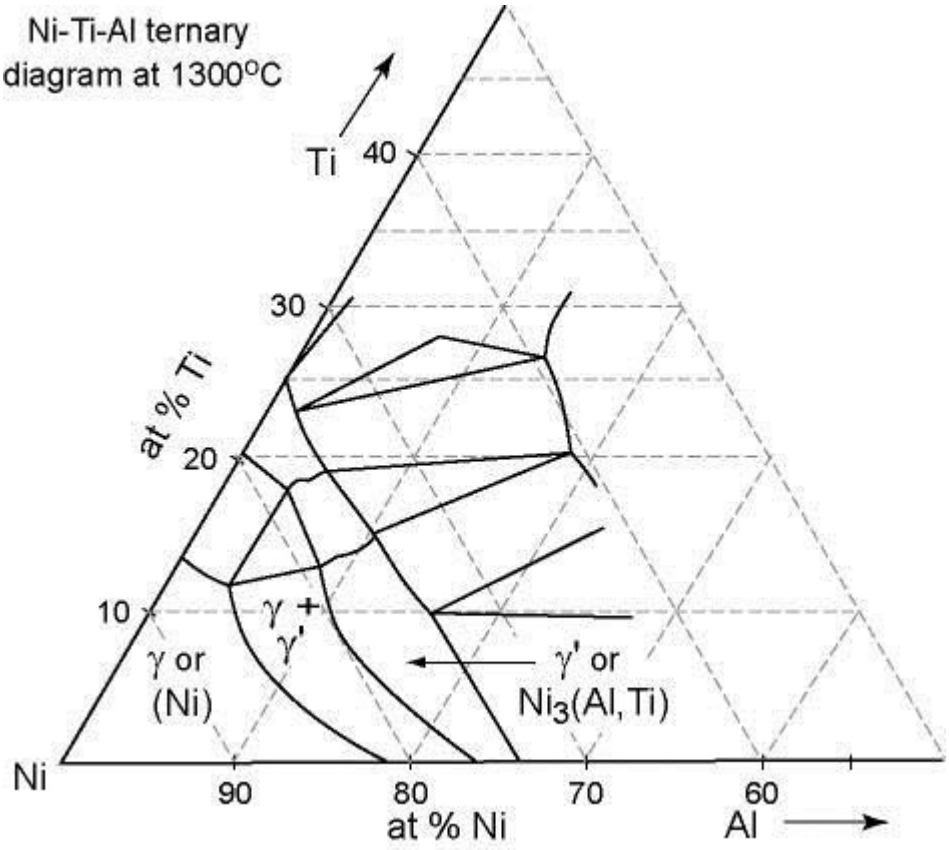
Design guidelines

Superalloys are expensive and, at room temperature, too hard to forge or roll; this can only be done hot. Those of highest strength are initially shaped by casting or by powder methods such as hot isostatic pressing, then finished by machining or grinding to give the final tolerance. High performance turbine blades, often of complex shape like the one in the photograph, are investment-cast. Their high temperature performance is further enhanced by causing the casting to solidify directionally, giving large oriented grains or a single crystal, increasing its resistance to creep.

Technical notes

All superalloys are age-hardened -- heated to dissolve the alloying elements, quenched to trap them in solution and then reheated (aged) to make them precipitate as a fine dispersion of particles. The precipitates are intermetallics such as Ni3Al and Ni3Ti. It is these that give the strength. The chromium adds to this, and also imparts resistance to hot gasses by creating a surface film of Cr2O3.

Phase diagram



Phase diagram description

Nickel based superalloys are alloys of nickel with aluminum, titanium and chromium. This is the nickel-rich corner of the nickel (Ni), aluminum (Al) titanium (Ti) phase diagram.

Typical uses

Blades, disks, and combustion chambers in turbines and jet engines, rocket engines, general structural aerospace applications, light springs, high temperature chemical engineering equipment, bioengineering and medical.

Tradenames

Inconel, Nimonic, Udimet, Haynes alloy, Hastalloy.

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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